

AGNES WANGECHI

Electrical & Electronics Engineer

AgnesWangechi.com | Engineeragneswangechi@gmail.com | +61 483 741 256 | Queensland, Australia | Open to full regional relocation | Seeking employer sponsorship

PROFESSIONAL SUMMARY

Electrical & Electronics Engineering graduate (University of Nairobi) with hands-on experience in telecommunications infrastructure, fibre-optic network deployment, and systems fault-finding at Safaricom PLC, Kenya's largest telco. Capstone project on a dual-axis solar tracker for optimised power generation. Strong grounding in network and systems engineering, root-cause analysis, and technical documentation, with exposure to Agile delivery and cross-functional coordination with DevOps and engineering teams. Seeking an entry- to mid-level role in power systems and control systems in Australia; open to regional relocation and employer sponsorship. Have a positive skill assessment from VETASSESS.

EDUCATION

Bachelor of Science in Electrical and Electronics Engineering

Sep 2017 – Dec 2022

University of Nairobi, Kenya

Capstone project: Design of a dual-axis solar tracker for optimising solar power efficiency.

TECHNICAL SKILLS

Programming Languages: TypeScript, JavaScript (ES2015+), Python, Dart, HTML, CSS, C++, C, MATLAB, SQL, Git.

Frameworks and Libraries: Tailwind CSS, Styled Components, Pandas, Matplotlib, Numpy, React, React Native, Django, Flask, Nodejs, REST APIs, Flutter, GraphQL

Tools & Platforms: Git, GitHub, Docker, AWS Amplify, Firebase, Vercel, Netlify, Figma, WordPress, MongoDB, Jupyter Notebook, Notion, Slack, Jira, Linux, Kubernetes, Raspberry Pi, Arduino, MySQL, Postman,

Networking & Telecommunications: TCP/IP, fibre-optic network infrastructure, routing protocols, firewall configuration, VLAN configuration, 5G network rollout, network security, fault finding

Systems Engineering & Operations: Root-cause analysis, fault-finding, system handovers, production incident investigation, technical requirements review, change management

Project delivery: Agile delivery, technical documentation, quantitative & qualitative data analysis, stakeholder coordination

Other: Valid driver's licence

PROFESSIONAL EXPERIENCE

Program Co-ordinator

Jun 2025 – Present

International Education Research Project, University of Nairobi (funded by CODE, in partnership with Canada) | Nairobi, Kenya

- Oversee administration of data-collection surveys, key informant interviews, focus groups, classroom observations, and document analysis across multiple field sites.
- Manage data organisation, transcription oversight, cleaning, and secure storage to maintain research integrity and audit readiness.

- Support quantitative and qualitative data preparation using descriptive, inferential, and thematic analysis approaches.
- Coordinate research logistics, travel, field permissions, and stakeholder communication with schools and partner bodies.

Billing Engineer (Product Developer)

Aug 2023 – Jun 2025

Safaricom PLC | Nairobi, Kenya

Supported business-critical **Telecommunications Business Support Systems (BSS)**, specifically Safaricom's **Convergent Billing System (CBS)**, covering billing operations, production support, system enhancements, data analysis, testing and incident resolution.

- Supported production operations for Safaricom's **Telecommunications BSS/CBS environment**, supporting approximately **40 million active customers** across high-volume Voice, SMS and Okoa services.
- Supported the development and maintenance of the **Okoa Chap Chap CBS product**, contributing to approximately **KES 120 million in revenue within three months** while supporting a large-scale customer base.
- Monitored and analysed **BSS/CBS system performance, alerts, charging data and production incidents**, identifying anomalies and potential service-impacting issues and coordinating investigation and resolution.
- Analysed **CBS Charging Data Records (CDRs)** using **SQL and Excel** to investigate billing and charging anomalies, data inconsistencies, system behaviour and service issues.
- Supported **incident triage, classification and prioritisation of P1, P2 and P3 production incidents**, coordinating investigation and escalation with Backend, DevOps, Operations and vendor teams.
- Investigated **P1/P2/P3 production incidents affecting BSS billing and charging services**, performing root-cause analysis and implementing corrective actions that contributed to a **30% reduction in critical defects**.
- Supported **UAT and regression testing** for CBS enhancements, configuration changes, product releases and regulatory requirements before production deployment.
- Performed **billing reconciliation, data validation and variance analysis**, identifying discrepancies between expected and actual billing/charging results and supporting revenue assurance activities.
- Used **SQL and Excel** to analyse billing and BSS data, identify trends and anomalies, and produce operational and management insights for stakeholders.
- Reviewed and refined **functional and technical requirements** for BSS/CBS changes, ensuring alignment between business requirements and backend system capabilities.
- Supported **BSS configuration, change requests and controlled releases**, assessing potential impacts on production billing and charging services.
- Maintained **system support documentation, technical documentation, incident records, change requests and configuration records** to support operational continuity and controlled releases.
- Collaborated with **Backend, DevOps, Product, Operations, Finance, Compliance and vendor teams** to resolve complex billing and system issues.
- Coordinated with external **vendors and contractors** during incident investigations, system enhancements, testing and deployment activities.

- Provided **data-driven system performance recommendations** to senior stakeholders and contributed to continuous improvement of billing processes and system reliability.
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MAJOR PROJECT

Huawei BSS/CBS Billing System Migration & 24/7 Production Support

Safaricom PLC | 12-month enterprise migration project

Participated in the **enterprise migration and upgrade of the telecommunications billing platform to Huawei**, supporting the project alongside normal Billing Engineer responsibilities for approximately one year before transitioning to dedicated migration and production-support activities during the final two months.

- Participated in a **12-month BSS/CBS billing-system migration project**, supporting technical, operational and business activities throughout the migration lifecycle.
- Transitioned from normal day-to-day responsibilities to **dedicated migration and production-support activities during the final two months**, supporting **24/7 system operations** during the critical migration and cutover period.
- Provided **24/7 production support** during migration, cutover and post-migration stabilisation, supporting monitoring, incident triage, escalation and resolution of service-impacting issues.
- Supported **P1/P2/P3 incident classification and escalation**, ensuring critical billing and charging incidents were prioritised and communicated to appropriate technical teams.
- Monitored **system alerts, performance metrics, database activity and technical logs** to identify anomalies and potential service disruptions.
- Investigated complex **BSS/CBS billing, charging and production incidents**, coordinating resolution with Huawei, Backend, DevOps, Operations and other technical teams.
- Performed **SQL-based data validation and analysis** to verify migrated billing data and investigate discrepancies between legacy and target systems.
- Supported **billing reconciliation, migration validation, UAT and regression testing** to verify billing accuracy and system functionality before and after migration.
- Participated in **cutover, post-migration stabilisation and hypercare**, supporting rapid identification and resolution of issues affecting critical billing services.
- Supported cross-functional **migration war-room activities**, coordinating with technical teams, vendors and business stakeholders during high-priority incidents.
- Maintained incident records, technical documentation, configuration information and **handover documentation** to support continuity of 24/7 operations.
- Contributed to ensuring **system reliability and service continuity** throughout the migration of a critical telecommunications billing platform.

Telecommunications Engineer

Mar 2023 – Jul 2023

Safaricom PLC | Nairobi, Kenya

- Implemented new fibre-optic network infrastructure deployment and technical configuration, resulting in 2,000 new fibre subscribers within 2 months of installation.
- Managed on-site field technicians, enforcing strict industrial workplace health and safety (WHS) protocols while ensuring deliverables met rigid timelines.
- Supervised the physical installation and technical optimization of industrial fiber-optic networks and backup radio links to secure communication between field equipment and operations centers.
- Provided technical support and troubleshooting for customer issues, resolving 95% within SLA timeframes.
- Supported the hardware testing, pre-commissioning, pilot roll-out and full live cutover of 5G network infrastructure across 3 regions in Nairobi.
- Gained hands-on experience with routing protocols, firewall settings, and VLAN configurations.

PROFESSIONAL RECOGNITION

VETASSESS skills assessment: Positive skill assessment in 511112: Program or Project Coordinator/Administrator.

REFERENCES

Available upon request.

